

Bijan Pourriahi

Senior Backend / Platform Engineer

Rails · PostgreSQL · Python · Rust · AWS · Linux

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PROFILE

Senior full-stack / platform engineer with 10+ years of experience turning complex business workflows into reliable production software. Built SaaS products, market research platforms, hedge-fund automation, analytics systems, internal tools, dashboards, deployment workflows, and data pipelines across startups, agencies, SaaS companies, and finance.

Strongest in roles where software has to make operations faster, clearer, and easier to run: clarifying business workflows, designing APIs and software architecture, leading engineers, managing deployments, automating manual work, improving system visibility, and keeping production systems maintainable.

SELECTED IMPACT

- Built proprietary financial systems infrastructure for research, simulation, account monitoring, broker integrations, risk controls, dashboards, data pipelines, deployment automation, and production observability.
- Designed high-speed simulation workflows capable of evaluating 1,000+ strategy/configuration variants and running year-long 1-minute-bar backtests in roughly 1-2 seconds.
- Led post-acquisition integration of five acquired SaaS systems at FocusVision, coordinating engineers and sysadmin teams across data migrations, infrastructure changes, production releases, and customer-impacting fixes.
- Owned end-to-end delivery of production software in small teams, including architecture, APIs, Rails applications, PostgreSQL data models, deployments, monitoring, incident response, and internal tools.
- Improved Solr/search indexing performance by roughly 100x by replacing a slow indexing workflow with a custom Java pipeline.

CORE SKILLS

Engineering leadership: software architecture, API design, team leadership, deployment ownership, production operations, cross-team coordination.

Product/platform delivery: SaaS features, admin tools, dashboards, data migrations, monitoring, automation, internal tools, customer-facing workflows.

Data and infrastructure: data modeling, analytics pipelines, search/indexing, performance tuning, DevOps, CI/CD, Linux, AWS, Kubernetes, Docker.

PROFESSIONAL EXPERIENCE

QuantBox — Principal Engineer

2021-Present | Remote

- **Context:** Product/research platform for data-heavy automation, market research, and operator workflows.
- Built an end-to-end research and operations platform that let one operator evaluate, deploy, monitor, and control 11 live accounts from one system instead of juggling scripts, broker tools, spreadsheets, screenshots, and manual review.
- Designed APIs, data models, and runtime architecture for a DB-backed control plane used to manage live execution, broker integrations, account state, risk controls, monitoring, and operator actions.
- Built high-speed backtesting and research workflows that evaluated 1,000+ strategies/configurations and ran 1-minute-bar year backtests in roughly 1-2 seconds.
- Built production/staging infrastructure with daily-or-better deployments, binary release bundles, health checks, failure alerts, Grafana/Prometheus monitoring, and Postgres-to-ClickHouse replication.
- Integrated XGBoost-based classifiers, replayable evidence workflows, and operator dashboards to support faster strategy comparison, disciplined deployment decisions, live-system review, and reliable daily operation.

Rare Isle Fund — Principal Software Engineer

Apr 2019-Jun 2021 | Remote

- **Context:** Small hedge fund; principal engineer responsible for core automation, analytics, dashboards, and production operations.
- Built automated trading, research, analytics, and operations infrastructure from scratch, replacing manual research and strategy review with repeatable software workflows.
- Designed large-scale backtesting and strategy evaluation pipelines capable of processing millions of strategy variants daily, helping the fund test and compare ideas faster.
- Built live trading systems, automated selection workflows, analytics dashboards, operational controls, and monitoring tools that improved visibility into deployed systems, fund resources, and executed trades.
- Owned implementation and operations across Python, Pandas, Go, AWS, Grafana, Linux, DevOps, data pipelines, and production infrastructure.

FocusVision — Senior Software Engineer

Nov 2015-Jun 2018 | Portland, OR

- **Context:** Market research SaaS company; acquired product environment with multiple systems and integration needs.
- Led engineering work to integrate five acquired companies/systems into FocusVision, consolidating multiple Rails/PostgreSQL systems into a more unified production platform for customer and internal workflows.
- Ran and mentored engineering teams while delivering SaaS features, integration work, bug fixes, and production improvements.
- Managed deployments and coordinated with sysadmin/operations teams across production releases, infrastructure changes, and integration work.
- Built and maintained customer-facing features, admin tools, APIs, database-backed workflows, internal operations systems, deployment automation, and release processes.
- Owned architecture and delivery work across data migration, production support, release workflows, and cross-team technical coordination.

ADDITIONAL EXPERIENCE

24tru — Software Engineer

Apr 2013-Jun 2015 | Lakewood, CO

- **Context:** Small market research SaaS team building software for research collection and video-based workflows.
- Co-owned the company's software platform with one other developer, covering production maintenance, data migrations, new features, customer-impacting fixes, and day-to-day technical operations.
- Managed deployment of the company's software stack from the ground up, including release processes, production maintenance, and operational reliability.
- Improved search/Solr indexing performance by roughly 100x by building a custom Java indexing application.
- Built APIs and product features that helped customers collect, upload, process, transcribe, search, and manage video research data.
- Managed data migrations, backend maintenance, deployment automation, DevOps, production support, and release reliability in a two-developer engineering team.

Tango Group — Software Developer

Feb 2012-Oct 2012 | Boulder, CO

- **Context:** Digital agency/client-services environment.
- Implemented Ruby on Rails features, fixed bugs, wrote tests, and supported PostgreSQL-backed client product development.

TECHNICAL STACK

Languages: Rust, Python, Ruby, Go, TypeScript, SQL

Frontend: React, TypeScript, HTML, CSS, dashboards, browser extensions, admin tools

Backend: Ruby on Rails, FastAPI, APIs, WebSockets, background jobs, service architecture, CLI systems

Data: PostgreSQL, ClickHouse, SQLite, ETL, data modeling, event logs, analytics pipelines

Infrastructure: Kubernetes, Docker, AWS, Linux, Ansible, CI/CD, sysadmin, Grafana, monitoring

Domains: SaaS, internal tools, automation, analytics, research infrastructure, trading systems, data-heavy products

EDUCATION

University of Colorado Boulder — B.S. Computer Science, 2012